

COMMAND LINE CRASH COURSE

Master of Science in Machine Learning and Data Science (MLDS)
Class of 2027, Omicron Cohort

Agenda

1. Key Terms

- Command Line vs Terminal vs Shell

2. Basics Commands

3. Common Use Cases

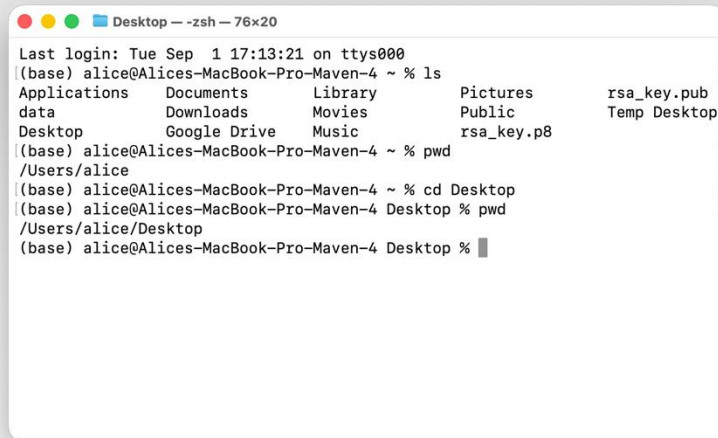
- Practical Applications
- Data Wrangling

4. Next Steps

KEY TERMS

What is the Command Line?

Term	Description
Command line / CLI	Command line interface = typing text commands instead of clicking via a graphical user interface (GUI)

A terminal window titled "Desktop -- zsh -- 76x20" showing a series of commands and their outputs. The user 'alice' is logged in on 'Alices-MacBook-Pro-Maven-4'. The terminal shows the execution of 'ls', 'pwd', 'cd Desktop', and another 'pwd' command, demonstrating basic command line navigation.

```
Desktop -- zsh -- 76x20
Last login: Tue Sep  1 17:13:21 on ttys000
(base) alice@Alices-MacBook-Pro-Maven-4 ~ % ls
Applications  Documents      Library        Pictures      rsa_key.pub
data          Downloads     Movies         Public        Temp Desktop
Desktop       Google Drive  Music          rsa_key.p8
(base) alice@Alices-MacBook-Pro-Maven-4 ~ % pwd
/Users/alice
(base) alice@Alices-MacBook-Pro-Maven-4 ~ % cd Desktop
(base) alice@Alices-MacBook-Pro-Maven-4 Desktop % pwd
/Users/alice/Desktop
(base) alice@Alices-MacBook-Pro-Maven-4 Desktop %
```

What is the Command Line?

Term	Description
Command line / CLI	Command line interface = typing text commands instead of clicking via a graphical user interface (GUI)
Terminal Apps	The application you open to access the command line • Mac: Terminal • Windows: cmd (old default), PowerShell (modern, powerful), Git Bash (commands match Mac)



What is a terminal?

In the 1960s, computers would fit in an entire room and the way to access it was through a terminal aka a keyboard and screen that was connected to the central computer

What is the Command Line?

Term	Description
Command line / CLI	Command line interface = typing text commands instead of clicking via a graphical user interface (GUI)
Terminal Apps	The application you open to access the command line • Mac: Terminal • Windows: cmd (old default), PowerShell (modern, powerful), Git Bash (commands match Mac)
Shell / Interpreter	The program that interprets and executes your commands • On Mac Terminal / Git Bash – this is a separate program running inside the terminal app (zsh for Mac and bash for Git Bash) • For cmd and PowerShell – the app is the shell



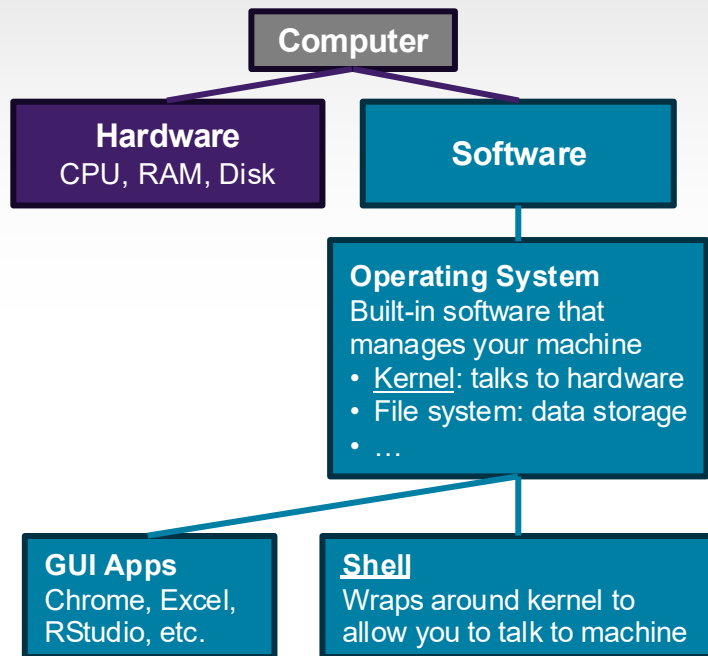
So which term should I use?

Just like people say R and RStudio interchangeably, you'll hear people say all these terms interchangeably – command line / terminal / shell

What is the Command Line?

Term	Description
Command line / CLI	Command line interface = typing text commands instead of clicking via a graphical user interface (GUI)
Terminal Apps	The application you open to access the command line • Mac: Terminal • Windows: cmd (old default), PowerShell (modern, powerful), Git Bash (commands match Mac)
Shell / Interpreter	The program that interprets and executes your commands • On Mac Terminal / Git Bash – this is a separate program running inside the terminal app (zsh for Mac and bash for Git Bash) • For cmd and PowerShell – the app is the shell

Why is it called the shell?



What is the Command Line?

Term	Description
Command line / CLI	Command line interface = typing text commands instead of clicking via a graphical user interface (GUI)
Terminal Apps	The application you open to access the command line • Mac: Terminal • Windows: cmd (old default), PowerShell (modern, powerful), Git Bash (commands match Mac)
Shell / Interpreter	The program that interprets and executes your commands • On Mac Terminal / Git Bash – this is a separate program running inside the terminal app (zsh for Mac and bash for Git Bash) • For cmd and PowerShell – the app is the shell

“command line”

How it *looks* (text-based vs GUI)

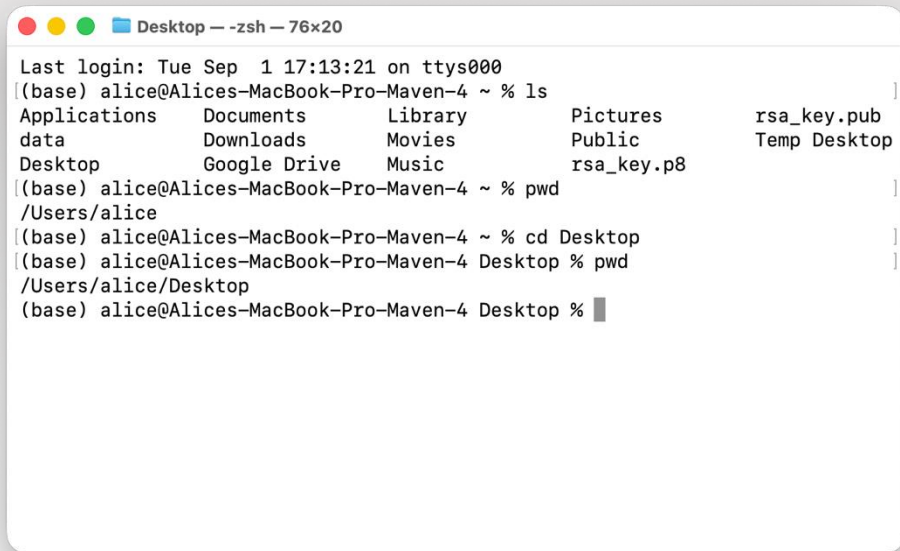
“terminal”

How it *accesses* the computer
(your window into the machine)

“shell”

How it *relates* to the computer
(the program that talks to the kernel)

What is the Command Line?



```
Desktop -- zsh -- 76x20
Last login: Tue Sep  1 17:13:21 on ttys000
(base) alice@Alices-MacBook-Pro-Maven-4 ~ % ls
Applications  Documents      Library        Pictures      rsa_key.pub
data          Downloads     Movies         Public        Temp Desktop
Desktop       Google Drive  Music          rsa_key.p8
(base) alice@Alices-MacBook-Pro-Maven-4 ~ % pwd
/Users/alice
(base) alice@Alices-MacBook-Pro-Maven-4 ~ % cd Desktop
(base) alice@Alices-MacBook-Pro-Maven-4 Desktop % pwd
/Users/alice/Desktop
(base) alice@Alices-MacBook-Pro-Maven-4 Desktop %
```

“command line”

How it *looks* (text-based vs GUI)

“terminal”

How it *accesses* the computer
(your window into the machine)

“shell”

How it *relates* to the computer
(the program that talks to the kernel)

“how you can talk directly with your machine”

Why Learn Command Line?

- Essential for git
 - git's real interface is command line – GUIs are just wrappers
- Essential as a data scientist
 - installs, configurations, environments, etc. – need to be proficient
 - data wrangling – sometimes it's easier / faster here (*examples to follow!*)
- Gets you closest to your computer
 - you are talking directly to your machine and can have full control

BASIC COMMANDS

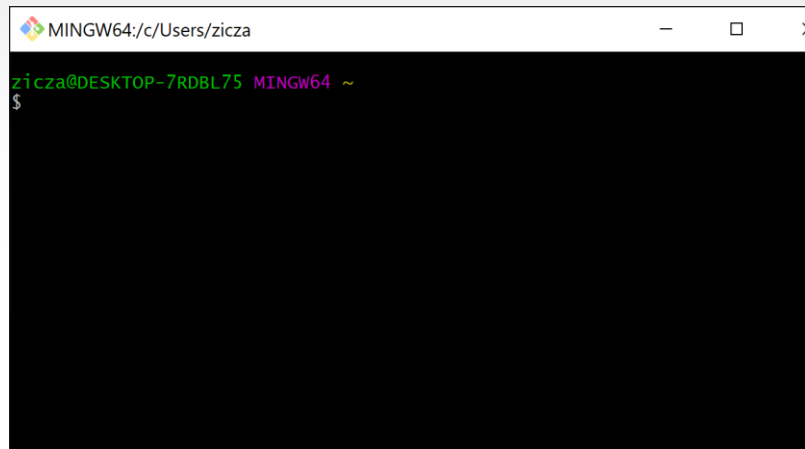
Getting Started

MacOS: Terminal

A screenshot of a macOS Terminal window. The title bar shows three colored window control buttons (red, yellow, green) followed by the text "alice -- zsh -- 76x20". The terminal content shows the prompt "(base) alice@Alices-MacBook-Pro-Maven-4 ~ %" with a cursor at the end of the line.

```
alice -- zsh -- 76x20
(base) alice@Alices-MacBook-Pro-Maven-4 ~ %
```

Windows: Git Bash

A screenshot of a Windows Git Bash terminal window. The title bar shows the Git logo followed by the text "MINGW64:/c/Users/zicza" and standard window control buttons. The terminal content shows the prompt "zicza@DESKTOP-7RDBL75 MINGW64 ~" with a cursor at the end of the line.

```
MINGW64:/c/Users/zicza
zicza@DESKTOP-7RDBL75 MINGW64 ~
$
```

Basic Commands & Shortcuts

Navigation

<code>pwd</code>	- print working directory
<code>ls / ls -la</code>	- list contents
<code>cd</code>	- change directory
<code>cd ~ / ..</code>	- go home / up a folder

Files & Folders

<code>mkdir</code>	- make a new directory
<code>touch</code>	- create an empty file
<code>cp</code>	- copy a file
<code>mv</code>	- move or rename a file
<code>rm / rm -r</code>	- remove a file / folder

Viewing Content

<code>cat</code>	- print the whole file
<code>head / tail / -n</code>	- print first/last lines

Shortcuts: Terminal

<code>↑ / ↓</code>	- view previous commands
<code>Tab</code>	- autocomplete
<code>Ctrl + C</code>	- cancel current command
<code>clear</code>	- clear the window

Shortcuts: Cursor

<code>← / →</code>	- move cursor left/right
<code>Option/Ctrl + ← →</code>	- jump cursor by word
<code>Ctrl + A / Ctrl + E</code>	- jump to start / end
<code>Ctrl + U</code>	- clear the whole line
<code>Ctrl + K</code>	- clear from cursor to end

Shortcuts: New

<code>Ctrl + T / N</code>	- Terminal: new tab / window
<code>Ctrl + Shift + N</code>	- Git Bash: new window

DEMO: Basic Commands & Shortcuts

1. Navigation

- Explore your home, Desktop and Downloads folders

2. Files & Folders – Create

- Create a new folder and an empty file inside
- Enter data into the file via GUI (Text Edit / Notepad)
- View the file and contents of the files

3. Files & Folders – Move

- Copy, rename and move things around
- Clean up files and folders

COMMON USE CASES

DEMO: Practical Applications

1. Checking installs

- `R --version`
- `python -version`

2. Launching apps

- `R / q()`
- `python / quit()`

3. Installing apps

- `brew install <app>` (MacOS)
- `winget install <app>` (Windows)

4. Working with vim

- `i` -- get into insert mode
- `Esc` -- get out of insert mode
- `:wq` -- write (aka save) and quit

DEMO: Data Wrangling

```
curl -O https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv  
wc -l titanic.csv  
  
grep -i "ann" titanic.csv  
  
grep -ci "ann" titanic.csv  
  
grep -E "\bAnn\b" titanic.csv  
  
grep -cE "\bAnn\b" titanic.csv  
  
grep -E "\bAnn\b" titanic.csv > ann_only.csv  
  
cat ann_only.csv
```

NEXT STEPS

COMMAND LINE CRASH COURSE

SUMMARY

-  The command line is just another way of talking to your computer
-  Command line / terminal / shell / bash = used interchangeably
-  You now know ~2/3 of the commands that a data scientist uses daily on the job – the rest will come from hands-on practice in your courses

NEXT STEPS

-  Don't memorize — know what's possible, look it up when you need it
-  Build muscle memory through repetition — use it daily and it'll stick

SMALL GROUP DISCUSSION

We've completed our command line crash course! 🎉

Discuss in groups of 2-4 people:

- What's something new you learned today?
- What's something else you want to try do in the command line?